

Algebraic Techniques

Week 2B

Overview

Surds

Key Ideas - Introduction to Surds

- Rules for Surds*
- Rationalise the Denominator*

Key Idea – Introduction to Surds

Knowing how to manipulate surds is vital for success in any calculus course. We will now look at the laws for manipulation of surds and undertake the practice needed to master the concepts involved.

A surd is the name given to a particular **irrational number of the form $\sqrt[n]{a}$ that cannot be simplified**. A surd is defined as the n th root of a rational number which, when expressed as a decimal, neither terminates nor recurs.

Definition:

$\sqrt[n]{a}$ is called a surd when a is not a perfect n^{th} power of any rational number.

Notes:

- $\sqrt{}$ is also referred to as a **radical sign**.
- We read the expression $\sqrt[n]{a}$ as **the n th root of a** .
- In particular, we read $\sqrt{}$ as **square root** and $\sqrt[3]{}$ as **cube root**.
- In the case when $n = 2$, the small n is omitted from the root sign. So, we read $\sqrt{a}$ as **the square root of a** .

Exercise: Evaluate each expression and determine if it is a surd:

(a) $\sqrt{0} = 0$

not a surd

(b) $\sqrt{1} = 1$

not a surd

(c) $7\sqrt{2}$

surd

(d) $\sqrt{3}$

surd

(e) $\sqrt{4} = 2$

not a surd

(f) $\sqrt{6}$

surd

(g) $\sqrt{7}$

surd

(h) $\sqrt[3]{8} = 2$

not a surd

(i) $\sqrt{9} = 3$

not a surd

Key Idea – Rules for Surds

Discussion Question:

Which of the following statements are true?

$$\sqrt{4} \times \sqrt{9} = \sqrt{36}$$

$$\begin{aligned} 2 \times 3 &= 6 \\ 6 &= 6 \end{aligned} \quad \text{true}$$

$$\sqrt{\frac{16}{4}} = \frac{\sqrt{16}}{\sqrt{4}}$$

$$\begin{aligned} \sqrt{4} &= \frac{4}{2} \\ 2 &= 2 \end{aligned} \quad \text{true}$$

$$(\sqrt{25})^2 = 25$$

$$\begin{aligned} 5^2 &= 25 \\ 25 &= 25 \end{aligned} \quad \text{true}$$

Rules involving the four basic arithmetic operations are used to simplify expressions involving surds.

Rules for Surds: If a and b are positive numbers, then:

1. $\sqrt{xy} = \sqrt{x} \times \sqrt{y}$
2. $\sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}}$
3. $(\sqrt{x})^2 = x$, provided $x \geq 0$
4. Only *like* surds can be added and subtracted.

Note: 1. $\sqrt{x} \pm \sqrt{y} \neq \sqrt{x \pm y}$

2. $\sqrt{x^2} = |x|$

3. We cannot simplify the expression $\sqrt{x^2 + y^2}$, i.e., $\sqrt{x^2 + y^2} \neq x + y$!!!

Before we look at each of the above rules, we need to be able to simplify surds, as this skill will be needed whilst using the rules.

Simplification of Surds

To manipulate surds properly, we need to be able to express them in their simplest form. A surd is expressed in its **simplest form** when the value of the number under the radical is **as small as possible**. When simplifying a surd, factorise the number under the radical sign, ensuring that one of the factors is the **largest perfect square possible**.

Recall perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121,...

Note:

A systematic way to simplify a surd is to divide the number in the surd by the largest, logical square number and work down from that.

Examples:

$$\begin{aligned}\sqrt{8} &= \sqrt{4 \times 2} \\ &= \sqrt{4} \times \sqrt{2} \\ &= 2 \times \sqrt{2} \\ &= 2\sqrt{2}\end{aligned}$$

$$\begin{aligned}2\sqrt{48} &= 2 \times \sqrt{16 \times 3} \\ &= 2 \times \sqrt{16} \times \sqrt{3} \\ &= 2 \times 4 \times \sqrt{3} \\ &= 8\sqrt{3}\end{aligned}$$

$$\begin{aligned}\sqrt{320} &= \sqrt{64 \times 5} \\ &= \sqrt{64} \times \sqrt{5} \\ &= 8 \times \sqrt{5} \\ &= 8\sqrt{5}\end{aligned}$$

Addition and Subtraction of Surds

As with algebraic terms, only like surds can be **added and subtracted**.

Examples:

The surd $\sqrt{2} + \sqrt{3}$ cannot be written in a simpler form as the two surds are **unlike surds**
(in the same way $2x$ and $3y$ are unlike terms in algebra)

The surd $5\sqrt{7} + 3\sqrt{7}$ can be written in a simpler form as the two surds are **like surds**

$$5\sqrt{7} + 3\sqrt{7} = 8\sqrt{7}$$

Before attempting to add or subtract surds, each surd should be **reduced to its simplest form by removing any factor that is a perfect square**.

Note: $\sqrt{x \pm y} \neq \sqrt{x} \pm \sqrt{y}$

Examples:

$$5\sqrt{3} + \sqrt{3} - 2\sqrt{3} = 4\sqrt{3}$$

$$\begin{aligned}\sqrt{27} + \sqrt{125} - \sqrt{20} - 2\sqrt{3} &= \sqrt{9} \times \sqrt{3} + \sqrt{25} \times \sqrt{5} - \sqrt{4} \times \sqrt{5} - 2\sqrt{3} \\ &= 3\sqrt{3} + 5\sqrt{5} - 2\sqrt{5} - 2\sqrt{3} \\ &= \sqrt{3} + 3\sqrt{5}\end{aligned}$$

$$\begin{aligned}\sqrt{27} + \sqrt{108} - \frac{1}{2}\sqrt{12} &= \sqrt{9} \times \sqrt{3} + \sqrt{36} \times \sqrt{3} - \frac{1}{2}\sqrt{4} \times \sqrt{3} \\ &= 3\sqrt{3} + 6\sqrt{3} - \sqrt{3} \\ &= 8\sqrt{3}\end{aligned}$$

$$\begin{aligned}5\sqrt{a} - 3\sqrt{b} - 3\sqrt{4a} &= 5\sqrt{a} - 3\sqrt{b} - 3 \times 2\sqrt{a} \\ &= 5\sqrt{a} - 3\sqrt{b} - 6\sqrt{a} \\ &= -\sqrt{a} - 3\sqrt{b}\end{aligned}$$

Exercises: Simplify the following surds:

$$\begin{aligned} \text{(a)} \quad 2\sqrt{63} + 0.5\sqrt{7} - \sqrt{112} &= 2 \times \sqrt{9 \times 7} + 0.5 \times \sqrt{7} - \sqrt{16 \times 7} \\ &= 2 \times \sqrt{9} \times \sqrt{7} + 0.5 \times \sqrt{7} - \sqrt{16} \times \sqrt{7} \\ &= 2 \times 3 \times \sqrt{7} + 0.5 \times \sqrt{7} - 4 \times \sqrt{7} \\ &= 6\sqrt{7} + 0.5\sqrt{7} - 4\sqrt{7} \\ &= 2.5\sqrt{7} \quad \text{or} \quad \frac{5\sqrt{7}}{2} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad 3\sqrt{x} - 3\sqrt{y} - 3\sqrt{9x} &= 3\sqrt{x} - 3\sqrt{y} - 3 \times \sqrt{9} \times \sqrt{x} \\ &= 3\sqrt{x} - 3\sqrt{y} - 3 \times 3 \times \sqrt{x} \\ &= 3\sqrt{x} - 3\sqrt{y} - 9\sqrt{x} \\ &= -6\sqrt{x} - 3\sqrt{y} \end{aligned}$$

Multiplication or Division of Surds

When **multiplying surds** :

- multiply the numbers outside the square root together
- multiply the numbers inside the square root together
- simplify if possible

A similar procedure holds for division, this process is outlined in the 1st and 2nd rules for surds:

$$a\sqrt{x} \times b\sqrt{y} = ab\sqrt{xy}$$

$$\frac{a\sqrt{x}}{b\sqrt{y}} = \frac{a}{b} \sqrt{\frac{x}{y}}$$

The 3rd rule, which is,

$$\left(\sqrt{x}\right)^2 = x, \text{ is a result of multiplication.}$$

The distributive law, $a(b + c) = ab + ac$, also holds when variables are replaced by surds when **expanding an expression with a binomial factor**.

Examples: Simplify the following surds:

$$\sqrt{4} \times \sqrt{9} = \sqrt{36} = 6$$

$$2\sqrt{7} \times \sqrt{27} = 2 \times \sqrt{189} = 2 \times \sqrt{9 \times 21} = 2 \times \sqrt{9} \times \sqrt{21} = 2 \times 3 \times \sqrt{21} = 6\sqrt{21}$$

$$\frac{15\sqrt{35}}{5\sqrt{7}} = \frac{15}{5} \sqrt{\frac{35}{7}} = 3\sqrt{5} \quad \text{Note this could be written as: } 15\sqrt{35} \div 5\sqrt{7} = \frac{15\sqrt{35}}{5\sqrt{7}} = \frac{15}{5} \sqrt{\frac{35}{7}} = 3\sqrt{5}$$

$$15\sqrt{2} \div 5\sqrt{8} = \frac{15\sqrt{2}}{5\sqrt{8}} = \frac{15}{5} \sqrt{\frac{2}{8}} = 3\sqrt{\frac{1}{4}} = 3\frac{\sqrt{1}}{\sqrt{4}} = \frac{3}{2}$$

$$\frac{\sqrt{20}}{\sqrt{64}} = \sqrt{\frac{20}{64}} = \frac{\sqrt{4 \times 5}}{8} = \frac{2\sqrt{5}}{8} = \frac{\sqrt{5}}{4}$$

$$\begin{aligned}
 (\sqrt{25})^2 &= \sqrt{25} \times \sqrt{25} \\
 &= \sqrt{125} \\
 &= 25
 \end{aligned}$$

$$(-2\sqrt{5})^2 = -2\sqrt{5} \times -2\sqrt{5} = 4 \times 5 = 20$$

$$\begin{aligned}
 (\sqrt{3} + \sqrt{5})^2 &= (\sqrt{3} + \sqrt{5}) \times (\sqrt{3} + \sqrt{5}) \\
 &= \sqrt{3}(\sqrt{3} + \sqrt{5}) + \sqrt{5}(\sqrt{3} + \sqrt{5}) \\
 &= 3 + \sqrt{15} + \sqrt{15} + 5 \\
 &= 8 + 2\sqrt{15}
 \end{aligned}$$

$$\begin{aligned}
 \sqrt{3}(\sqrt{3} + 5\sqrt{2}) &= \sqrt{3} \times \sqrt{3} + \sqrt{3} \times 5\sqrt{2} \\
 &= 3 + 5\sqrt{6}
 \end{aligned}$$

Exercises: Simplify the following surds.

$$\begin{aligned} \text{(a)} \quad 2\sqrt{2} \times 7\sqrt{6} &= 2 \times 7 \times \sqrt{2} \times \sqrt{6} \\ &= 14 \sqrt{2 \times 6} \\ &= 14 \sqrt{12} \\ &= 14 \times \sqrt{4} \times \sqrt{3} \\ &= 14 \times 2 \times \sqrt{3} \\ &= 28\sqrt{3} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \frac{2\sqrt{3}}{\sqrt{16}} &= \frac{2\sqrt{3}}{4} \\ &= \frac{\cancel{2}\sqrt{3}}{\cancel{2} \times 2} \\ &= \frac{\sqrt{3}}{2} \end{aligned}$$

Exercises: Simplify the following surds.

$$\begin{aligned} \text{(c)} \quad \sqrt{108} \div \sqrt{24} &= \frac{\sqrt{108}}{\sqrt{24}} \\ &= \frac{\sqrt{36} \times \sqrt{3}}{\sqrt{4} \times \sqrt{6}} \\ &= \frac{6\sqrt{3}}{2\sqrt{6}} \\ &= \frac{3 \times \cancel{2} \times \cancel{\sqrt{3}}}{\cancel{2} \times \cancel{\sqrt{3}} \times \sqrt{2}} = \frac{3}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \text{(d)} \quad (-3\sqrt{5})^2 &= (-3)^2 \times (\sqrt{5})^2 \\ &= 9 \times 5 = 45 \end{aligned}$$

Exercises: Expand and simplify:

$$\begin{aligned} \text{(a)} \quad (2\sqrt{6} - \sqrt{3})(\sqrt{6} + 3\sqrt{3}) &= 2\sqrt{6} \times \sqrt{6} + 2\sqrt{6} \times 3\sqrt{3} - \sqrt{3} \times \sqrt{6} - \sqrt{3} \times 3\sqrt{3} \\ &= 2(\sqrt{6})^2 + 6\sqrt{6 \times 3} - \sqrt{3 \times 6} - 3(\sqrt{3})^2 \\ &= 2 \times 6 + 5\sqrt{18} - 3 \times 3 \\ &= 12 + 5\sqrt{9 \times 2} - 9 \\ &= 3 + 5 \times \sqrt{9} \times \sqrt{2} \\ &= 3 + 5 \times 3 \times \sqrt{2} \\ &= 3 + 15\sqrt{2} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad (\sqrt{5} - \sqrt{3})(\sqrt{5} + \sqrt{3}) &= (\sqrt{5})^2 - (\sqrt{3})^2 \\ &= 5 - 3 \\ &= 2 \end{aligned}$$

difference of
two squares

Key Idea – Rationalising the Denominator

To simplify a fractional surd expression, we need to **rationalise the denominator**.

Rationalising the denominator is the process **of removing surds from the denominator of a fraction**.

There are two ways of rationalising, depending on the denominator.

1. For a fraction with a single surd as a factor on the denominator, **multiply numerator**

and denominator by the surd, for example $\frac{6}{5\sqrt{3}}$.

2. For a fraction with a **binomial denominator** (two terms in the denominator), at least one of which contains a surd, **multiply the numerator and denominator by the conjugate**

of the denominator, for example $\frac{\sqrt{5}}{3\sqrt{5}-\sqrt{2}}$ or $\frac{8}{\sqrt{3}+1}$.

Fractions with a Single Surd.

To rationalise the denominator of a fraction with a single surd, we need to recall:

$$\left(\sqrt{x}\right)^2 = \sqrt{x} \times \sqrt{x} = x$$

Examples: Rationalise the denominator:

$$\begin{aligned}\frac{4}{2\sqrt{5}} &= \frac{4}{2\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} \\ &= \frac{4\sqrt{5}}{10} \\ &= \frac{2\sqrt{5}}{5}\end{aligned}$$

$$\begin{aligned}\frac{2\sqrt{3}}{\sqrt{6}} &= \frac{2\sqrt{3}}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} \\ &= \frac{2\sqrt{18}}{6} \\ &= \frac{2 \times \sqrt{9} \times \sqrt{2}}{6} \\ &= \frac{6\sqrt{2}}{6} \\ &= \sqrt{2}\end{aligned}$$

Exercises: Rationalise the denominator of the fractional surd $\frac{5\sqrt{3}}{6\sqrt{15}}$.

$$\frac{5\sqrt{3}}{6\sqrt{15}} \times \frac{\sqrt{15}}{\sqrt{15}} = \frac{5\sqrt{3 \times 15}}{6 \times (\sqrt{15})^2}$$

$$= \frac{5\sqrt{45}}{6 \times 15}$$

$$= \frac{5\sqrt{9 \times 5}}{6 \times 15}$$

$$= \frac{\cancel{5} \times \sqrt{9} \times \sqrt{5}}{3 \times 2 \times 3 \times \cancel{5}}$$

$$= \frac{\cancel{3} \times \sqrt{5}}{\cancel{3} \times 2 \times 3} = \frac{\sqrt{5}}{6}$$

Fractions with a Binomial Denominator

Before we can rationalise the denominator of a fraction with two terms in the denominator, we need to understand the concept of conjugate expressions.

A conjugate expression is **formed by changing the sign between two terms in a binomial expression**. For example, the conjugate of $(x + y)$ is $(x - y)$. We can also say that $(x + y)$ is the conjugate of $(x - y)$. In other words, the two binomials are conjugates of each other. The expressions $(\sqrt{a} + \sqrt{b})$ and $(\sqrt{a} - \sqrt{b})$ are **conjugates** of each other. They form a conjugate pair.

When we multiply conjugate expressions by using the difference of two squares, the result is a rational number.

$$\begin{aligned}(\sqrt{a} + \sqrt{b}) \times (\sqrt{a} - \sqrt{b}) &= (\sqrt{a})^2 - (\sqrt{b})^2 \\ &= a - b\end{aligned}$$

Examples: Expand and simplify:

$$\begin{aligned}(4 - \sqrt{3})(4 + \sqrt{3}) &= 4^2 - (\sqrt{3})^2 \\ &= 16 - 3 \\ &= 13\end{aligned}$$

$$\begin{aligned}(\sqrt{11} + \sqrt{7})(\sqrt{11} - \sqrt{7}) &= (\sqrt{11})^2 - (\sqrt{7})^2 \\ &= 11 - 7 \\ &= 4\end{aligned}$$

$$\begin{aligned}\frac{3}{\sqrt{5} - 2} &= \frac{3}{\sqrt{5} - 2} \times \frac{\sqrt{5} + 2}{\sqrt{5} + 2} \\ &= \frac{3\sqrt{5} + 6}{(\sqrt{5})^2 - 2^2} \\ &= \frac{3\sqrt{5} + 6}{5 - 4} \\ &= 3\sqrt{5} + 6\end{aligned}$$

$$\begin{aligned}\frac{5}{7 - 2\sqrt{3}} &= \frac{5}{7 - 2\sqrt{3}} \times \frac{7 + 2\sqrt{3}}{7 + 2\sqrt{3}} \\ &= \frac{35 + 10\sqrt{3}}{7^2 - (2\sqrt{3})^2} \\ &= \frac{35 + 10\sqrt{3}}{49 - 12} \\ &= \frac{35 + 10\sqrt{3}}{37}\end{aligned}$$

Exercise:

Express the following surd with a rational denominator $\frac{8}{\sqrt{3}+1}$.

$$\begin{aligned}\frac{8}{(\sqrt{3}+1)} \times \frac{(\sqrt{3}-1)}{(\sqrt{3}-1)} &= \frac{8(\sqrt{3}-1)}{(\sqrt{3})^2 - 1^2} \\ &= \frac{8(\sqrt{3}-1)}{3-1} \\ &= \frac{8(\sqrt{3}-1)}{2} \\ &= 4(\sqrt{3}-1)\end{aligned}$$

Exercise: Evaluate the following, $\frac{2\sqrt{3}}{\sqrt{6}-\sqrt{3}} - \frac{\sqrt{3}}{2\sqrt{6}+3\sqrt{3}}$.

$$\frac{2\sqrt{3}(2\sqrt{6}+3\sqrt{3})}{(\sqrt{6}-\sqrt{3})(2\sqrt{6}+3\sqrt{3})} - \frac{\sqrt{3}(\sqrt{6}-\sqrt{3})}{(2\sqrt{6}+3\sqrt{3})(\sqrt{6}-\sqrt{3})}$$

$$= \frac{2\sqrt{3}(2\sqrt{6}+3\sqrt{3}) - \sqrt{3}(\sqrt{6}-\sqrt{3})}{(2\sqrt{6}+3\sqrt{3})(\sqrt{6}-\sqrt{3})}$$

$$= \frac{4\sqrt{18} + 6(\sqrt{3})^2 - \sqrt{18} + (\sqrt{3})^2}{2(\sqrt{6})^2 - 2\sqrt{6}\sqrt{3} + 3\sqrt{3}\sqrt{6} - 3(\sqrt{3})^2}$$

$$= \frac{3\sqrt{18} + 6 \times 3 + 3}{2 \times 6 + \sqrt{18} - 3 \times 3}$$

$$= \frac{3\sqrt{18} + 21}{3 + \sqrt{18}}$$

$$= \frac{3\sqrt{9 \times 2} + 21}{3 + \sqrt{9 \times 2}}$$

$$= \frac{3 \times \sqrt{9} \times \sqrt{2} + 21}{3 + \sqrt{9} \times \sqrt{2}}$$

$$= \frac{(9\sqrt{2} + 21) \times (3 - 3\sqrt{2})}{(3 + 3\sqrt{2}) \times (3 - 3\sqrt{2})}$$

$$= \frac{27\sqrt{2} - 27 \times (\sqrt{2})^2 + 63 - 63\sqrt{2}}{3^2 - (3\sqrt{2})^2}$$

$$= \frac{-36\sqrt{2} + 15}{-9}$$

$$= \frac{12\sqrt{2} - 5}{3}$$

Exercise: Evaluate the integers p and q , where $\frac{5+\sqrt{3}}{7+2\sqrt{3}} = p + q\sqrt{3}$.

$$\begin{aligned} \frac{(5+\sqrt{3}) \times (7-2\sqrt{3})}{(7+2\sqrt{3}) \times (7-2\sqrt{3})} &= \frac{35 - 10\sqrt{3} + 7\sqrt{3} - 2(\sqrt{3})^2}{7^2 - (2\sqrt{3})^2} \\ &= \frac{35 - 3\sqrt{3} - 2 \times 3}{49 - 4 \times 3} \\ &= \frac{29 - 3\sqrt{3}}{37} \\ &= \frac{29}{37} - \frac{3 \cdot \sqrt{3}}{37} \\ &= p + q\sqrt{3} \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} p = \frac{29}{37} \\ q = -\frac{3}{37} \end{array}$$

Exercise: If $x = 1 + \sqrt{3}$ and $y = \frac{1}{x}$, find the value of $x^2 + y^2$.

$$x^2 + y^2 = (1 + \sqrt{3})^2 + \left(\frac{1}{1 + \sqrt{3}}\right)^2$$

$$= 1 + 2\sqrt{3} + (\sqrt{3})^2 + \frac{1}{1 + 2\sqrt{3} + (\sqrt{3})^2}$$

$$= 4 + 2\sqrt{3} + \frac{1}{4 + 2\sqrt{3}}$$

$$= \frac{(4 + 2\sqrt{3})^2 + 1}{4 + 2\sqrt{3}}$$

$$= \frac{16 + 16\sqrt{3} + (2\sqrt{3})^2 + 1}{4 + 2\sqrt{3}}$$

$$= \frac{17 + 16\sqrt{3} + 4 \times 3}{4 + 2\sqrt{3}}$$

$$= \frac{(29 + 16\sqrt{3}) \times (4 - 2\sqrt{3})}{(4 + 2\sqrt{3}) \times (4 - 2\sqrt{3})}$$

$$= \frac{29 \times 4 - 29 \times 2\sqrt{3} + 16\sqrt{3} \times 4 - 16\sqrt{3} \times 2\sqrt{3}}{4^2 - (2\sqrt{3})^2}$$

$$= \frac{116 - 58\sqrt{3} + 64\sqrt{3} - 96}{16 - 4 \times 3}$$

$$= \frac{20 + 6\sqrt{3}}{4}$$

$$= \frac{2(10 + 3\sqrt{3})}{2 \times 2} = \frac{10 + 3\sqrt{3}}{2}$$